UNIVERSITY OF HERTFORDSHIRE
School of Engineering and Computer Science

PROJECT REPORT

Computer programming (4FTC2029)

Multiplayer Memory Game using Microcontroller

Submitted by: Group 1 (Team ID: 33)

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1.Introduction

1.1 Project Overview

This report documents the design, implementation, and validation of a Multiplayer Memory Game console built on an ATmega328P (Arduino Uno) microcontroller. The system is designed to challenge players through three distinct gameplay modes: Solo Sprint, Synchronous Duel, and The Challenger, that test players' memory, speed, and competitive ability through a sequence of LED flashes that must be replicated using physical pushbuttons, each targeting different cognitive and competitive skills.

The hardware includes eight push buttons, four shared LEDs, a 16x2 I2C LCD, a passive buzzer, and non-volatile storage to deliver a complete interactive gaming experience. The project is implemented entirely in register-level Embedded C, without any Arduino library abstractions, demonstrating core concepts in embedded systems programming including hardware timer configuration, interrupt service routines, pin change interrupts, EEPROM persistence, and power management.

Key software features include dynamic difficulty scaling, timeout management, persistent high-score storage, and power saving sleep mode. Audio feedback, real time LCD status updates, and competitive scoring logic enhance the user experience.

The system was tested across all game modes with documented pass/fail results, demonstrating reliable sequence detection, accurate timing, and correct score persistence across power cycles. The project demonstrates the practical application of embedded systems principles in the context of interactive entertainment, showcasing skills in hardware interfacing, real time programming, memory management, and human-machine interface design.

1.2 Objectives:

- Design and implement a hardware–software system centered on an ATmega328P microcontroller.
- All 3 game modes; Solo Sprint (single player), Sync Duel (two-player race), and The Challenger (two-player asymmetric). with difficulty scaling (shorter LED times + longer sequences)
- Persistent high scores via EEPROM (survives power cycles)
- Real-time LCD scoring during gameplay
- Implement a persistent leaderboard stored in EEPROM that survives power cycles.

The challenge is to integrate all these requirements into a reliable, responsive, and maintainable embedded firmware running on a resource-constrained 8-bit microcontroller.

2. Functionality

2.1 Hardware Inputs and Outputs

- 8 Push Buttons: 4 per player, mapped to LED indices 1–4 for sequence input.
- 4 Shared LEDs: Used for displaying randomly generated sequences.
- 16x2 I2C LCD: Displays menus, current mode, scores, and status messages.
- Passive Buzzer: Provides audio feedback on correct input, wrong input, and sequence playback.
- EEPROM/Flash: Stores high scores persistently across power cycles.
- Microcontroller — ATmega328P

The ATmega328P is an 8-bit AVR RISC microcontroller. Key resources used:
Memory: 32 KB Flash, 2 KB SRAM, 1 KB EEPROM.
Timers: Timer1 (16-bit) used for a 1 ms system tick.
I/O: Pin Change Interrupts (PCINT) on all GPIO banks.
Power: Power-Down sleep mode used for conservation.

2.2 Gameplay Modes

Mode 1; Solo Sprint: The system generates a random LED sequence of increasing length. The player must replicate the sequence by pressing the corresponding buttons. Upon correct input, the sequence length increments by one. An incorrect input or timeout terminates the round and displays the score.

Mode 2: Synchronous Duel: Both players observe the same LED sequence simultaneously. Both race to replicate it correctly; the faster correct responder scores one point. Multiple rounds accumulate to a final duel score.

Mode 3: The Challenger: Player A generate a secret LED sequence. Player B then attempts to reproduce it. If Player B succeeds, B scores a point. If B fails, Player A must reproduce the same sequence; success awards A a point, while failure awards no points. This mode introduces a strategic element of sequence complexity selection.

2.3 Functional Features

- Difficulty Scaling: LED display speed increases and sequence length grows proportionally with accumulated score.
- LCD Start Menu: Displays system title and a navigable mode-selection menu on startup.
- Audio Feedback: Distinct tones for correct input, incorrect input, sequence playback tones, and game-over notification.
- Timeout Management: A 5-second input timeout ends the current player turn with a failure.
- High Score Reset: A dedicated menu option allows the player to clear all stored high scores from EEPROM.

3. Hardware Design

3.1 Hardware Block Diagram

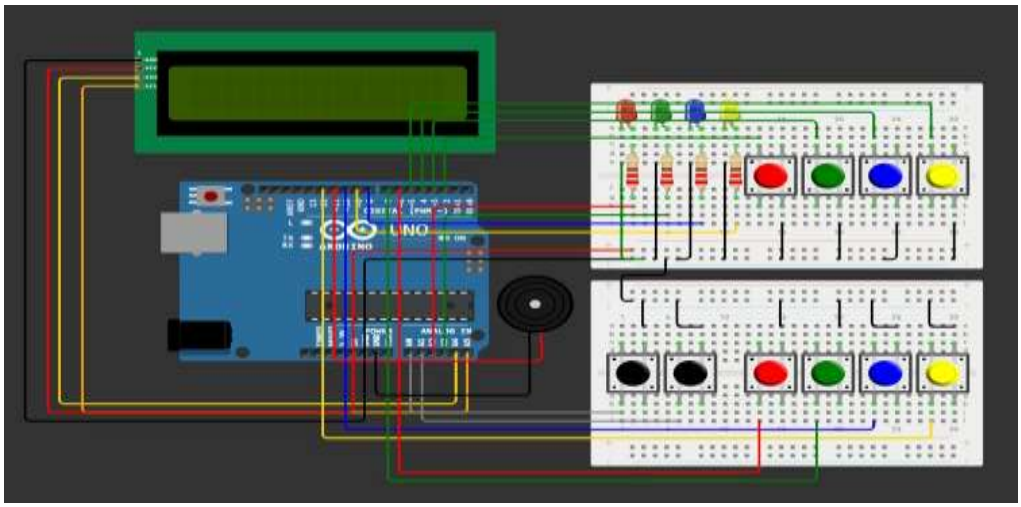

Figure 1:System Block Diagram

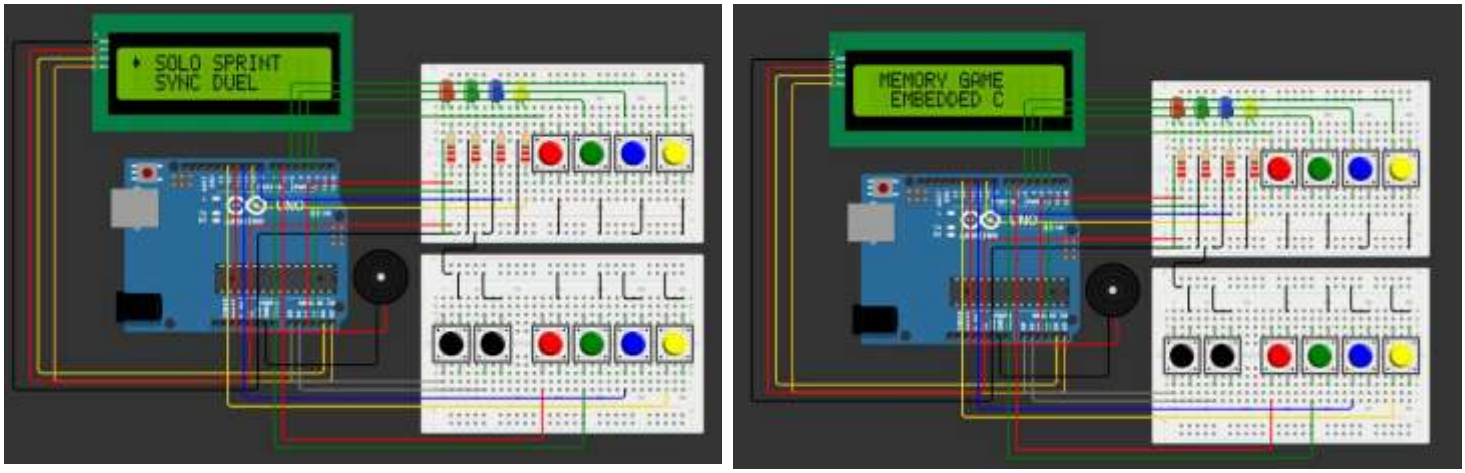

Figure 2:Full Circuit Schematic

3.2 Pin Assignment Table

Component	MCU Pin	Pin Interface	Direction	Description
Player 1 Button 1	PD 2	D2	Input (Pull-up)	Maps to LED 1
Player 1 Button 2	PD 3	D3	Input (Pull-up)	Maps to LED 2
Player 1 Button 3	PD 4	D4	Input (Pull-up)	Maps to LED 3
Player 1 Button 4	PD 5	D5	Input (Pull-up)	Maps to LED 4
Player 2 Button 1	PD 6	D6	Input (Pull-up)	Maps to LED 1
Player 2 Button 2	PD 7	D7	Input (Pull-up)	Maps to LED 2
Player 2 Button 3	PD 8	D8	Input (Pull-up)	Maps to LED 3
Player 2 Button 4	PD 9	D9	Input (Pull-up)	Maps to LED 4
LED 1 (Red)	PC 2	A2	Output – 220 Ω resistor	Sequence indicator 1
LED 4 (Green)	PC 4	A3	Output – 220 Ω resistor	Sequence indicator 2
LED 3 (Blue)	PC 5	D8	Output – 220 Ω resistor	Sequence indicator 3
LED 2 (Yellow)	PC 3	D9	Output – 220 Ω resistor	Sequence indicator 4
Nav Up/Select	PC0-PC1	A0- A1	Output (PWM)	Menu navigation
LCD (SDA)	PC 4	A4	I2C Data	16x2 I2C display
LCD (SCL)	PC 5	A5	I2C Clock	16x2 I2C display
EEPROM		Internal	Read/Write	ATmega328P built-in 1 KB

Table 1:Complete Hardware Pin Assignment

3.3 Circuit Architecture

All push buttons are wired in active-low configuration using the ATmega328P internal pull-up resistors (enabled via INPUT_PULLUP). This eliminates the need for external resistor arrays and reduces component count.

LEDs are driven through 220 Ohm current-limiting resistors to maintain approximately 10 mA forward current per LED.

The passive buzzer is driven via PWM on an analog-capable pin using the tone() library to generate distinct musical notes.

The I2C LCD requires only two signal wires (SDA and SCL) plus power, substantially reducing wiring complexity compared to a parallel interface.

3.4 Core Embedded Mechanisms

Timer1 CTC Mode (1 ms System Tick):
Timer1 is configured in CTC (Clear Timer on Compare Match) mode. With a prescaler of 64 and a CPU clock of 16 MHz, the OCR1A register is set to 249 to trigger an interrupt every 1 millisecond. This provides a non-blocking millis() function.

Pin Change Interrupts (PCINT):
The system uses PCINT to detect button presses. This allows the MCU to wake from Power-Down sleep. A critical fix implemented was using (1<<PC0) rather than symbolic PCINT names to avoid 8-bit register overflow.

EEPROM Persistence:
A "Magic Number" (0xA5) is stored at address 0x0000. If detected on boot, the system loads existing high scores; otherwise, it initializes the EEPROM. eeprom_update_word() is used to prevent unnecessary write cycles and extend memory life.

4. Software Design

4.1 Finite State Machine Architecture

The firmware is structured as a hierarchical Finite State Machine (FSM). Each state represents a distinct system condition, and transitions are triggered by validated events such as button presses, timer expiration, or completion flags. This architecture eliminates race conditions, makes behavior predictable, and facilitates independent testing of each state.

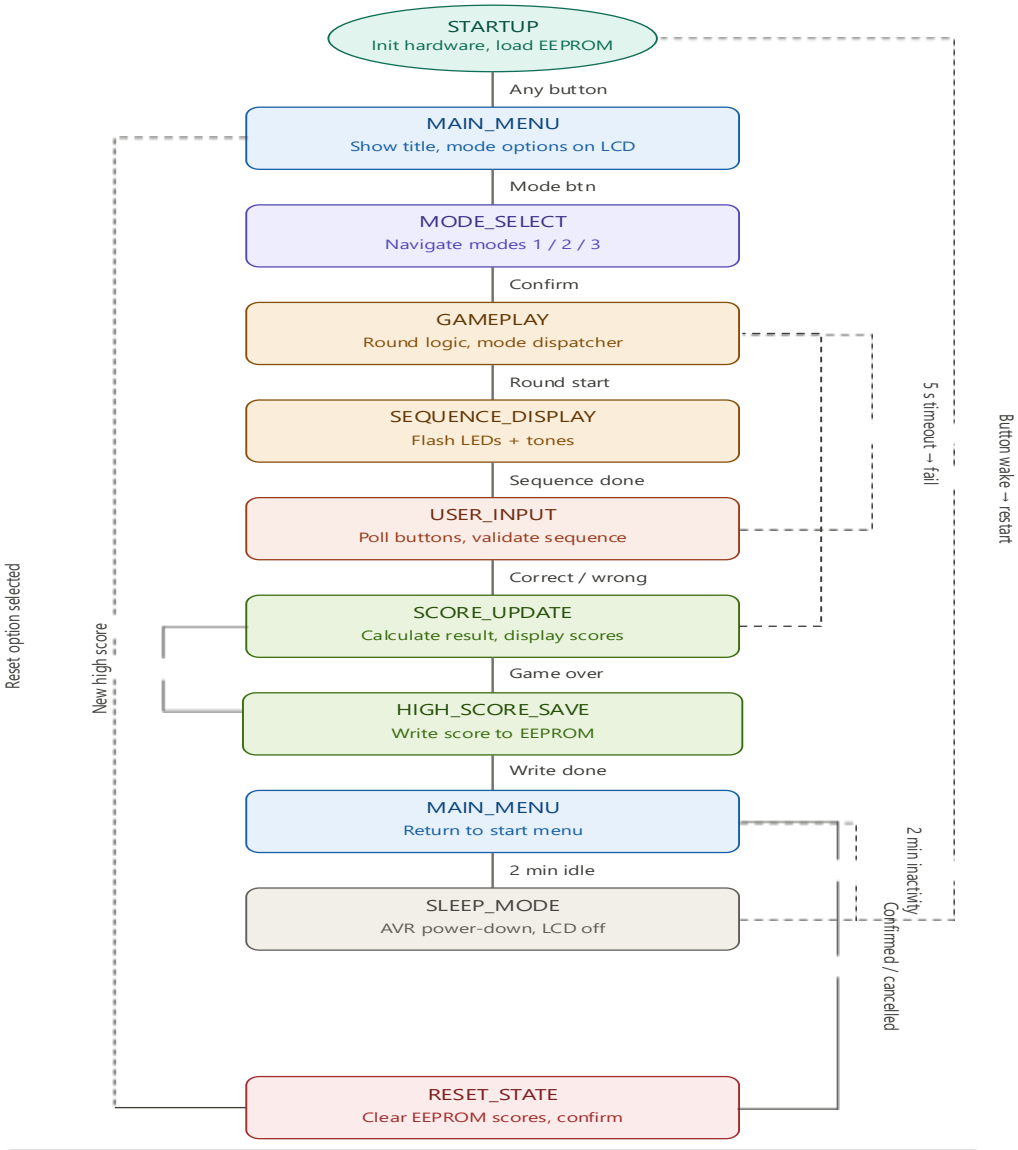

Figure 3:Finite State Machine Diagram

4.2 FSM State Descriptions

State	Description	Entry Condition	Exit Condition
STARTUP	Initialize hardware, load EEPROM high scores, display welcome splash	Power-on / Reset	Any button press
MAIN_MENU	Display game title and mode options on LCD	From STARTUP or SCORE_UPDATE	Mode selection button
MODE_SELECT	Navigate between Mode 1, 2, and 3	From MAIN_MENU	Confirm selection
GAMEPLAY	Manage round logic based on selected mode	From MODE_SELECT	Round complete or timeout
SEQ_DISPLAY	Flash LED sequence with tones	From GAMEPLAY	Sequence complete
USER_INPUT	Poll buttons, validate against sequence	From SEQ_DISPLAY	Correct / Wrong / Timeout
SCORE_UPDATE	Calculate and display round result, update scores	From USER_INPUT	After display delay
HIGH_SCORE_SAVE	Write updated scores to EEPROM	From SCORE_UPDATE (new high)	Write complete
SLEEP_MODE	Enter AVR power-down mode	2 min inactivity timer	External interrupt (button)
RESET_STATE	Clear EEPROM scores after confirmation	From MAIN_MENU option	Confirmed or cancelled

Table 2:FSM State Descriptions

4.4 Modular Software Architecture

Module	Responsibilities
input.h / input.cpp	Button polling, debouncing, player attribution
output.h / output.cpp	LED control, buzzer tone generation
display.h / display.cpp	LCD initialization, message rendering, menu navigation
game.h / game.cpp	Sequence generation, FSM transitions, game mode logic
memory.h / memory.cpp	EEPROM read/write, high score management
timer.h / timer.cpp	Non-blocking timer management via millis()
sleep.h / sleep.cpp	AVR sleep mode configuration and wake interrupt

Table 3:Software Module Breakdown

5.Complete implementation

5.1 Full Arduino Code

```
#ifndef CONFIG_H
#define CONFIG_H

#ifndef F_CPU
#  define F_CPU 16000000UL
#endif

#define BTN_P1_0_BIT   PD2
#define BTN_P1_1_BIT   PD3
#define BTN_P1_2_BIT   PD4
#define BTN_P1_3_BIT   PD5

#define BTN_P2_0_BIT   PD6
#define BTN_P2_1_BIT   PD7
#define BTN_P2_2_BIT   PB2
#define BTN_P2_3_BIT   PB4

#define BTN_NAV_UP_BIT   PC0
#define BTN_NAV_SEL_BIT  PC1

#define LED0_BIT   PC2
#define LED1_BIT   PC3
#define LED2_BIT   PB0
#define LED3_BIT   PB1

#define BUZZ_BIT   PB3

#define LCD_ADDR   0x27

#define EE_MAGIC   0
#define EE_HI1     2
#define EE_HI2     4
#define MAGIC      0xA5
```

```
#include <avr/io.h>
#include <avr/interrupt.h>
#include <avr/sleep.h>
#include <stdint.h>
#include <stdlib.h>
#include <stdio.h>
#include <string.h>

#include "config.h"
#include "lcd.h"
#include "input.h"
#include "game_common.h"
#include "buzzer.h"
#include "eeprom.h"
#include "mode1.h"
#include "mode2.h"
#include "mode3.h"

volatile uint32_t ms_count = 0;

static volatile uint32_t last_active = 0;
static          uint8_t  sleeping    = 0;

static int hi1 = 0;
static int hi2 = 0;

static uint8_t app_state = APP_SPLASH;
static uint8_t menu_sel  = 0;

static int p1_score = 0, p2_score = 0;

static uint8_t  reset_confirm = 0;
static uint32_t reset_tmr     = 0;
```

```c
#include <avr/io.h>
#include <avr/interrupt.h>
#include <stdlib.h>
#include <stdint.h>
#include "config.h"
#include "game_common.h"

uint32_t millis(void) {
    uint32_t t;
    uint8_t  sreg = SREG;
    cli();
    t = ms_count;
    SREG = sreg;
    return t;
}

void led_on(uint8_t i) {
    switch (i) {
        case 0: PORTC |= (1 << LED0_BIT); break;
        case 1: PORTC |= (1 << LED1_BIT); break;
        case 2: PORTB |= (1 << LED2_BIT); break;
        case 3: PORTB |= (1 << LED3_BIT); break;
        default: break;
    }
}

void led_off(uint8_t i) {
    switch (i) {
        case 0: PORTC &= (uint8_t)~(1 << LED0_BIT); break;
        case 1: PORTC &= (uint8_t)~(1 << LED1_BIT); break;
        case 2: PORTB &= (uint8_t)~(1 << LED2_BIT); break;
        case 3: PORTB &= (uint8_t)~(1 << LED3_BIT); break;
        default: break;
    }
}
```

```c
#ifndef GAME_COMMON_H
#define GAME_COMMON_H

#include <stdint.h>
#include "config.h"

extern volatile uint32_t ms_count;
uint32_t millis(void);

void led_on(uint8_t idx);
void led_off(uint8_t idx);
void leds_off(void);

void seq_make(int *seq, int len);

void    flash_start(int *seq, int len, int on_ms);
uint8_t flash_run(void);
int     flash_ms(int round);

#endif

```

```
#ifndef EEPROM_H
#define EEPROM_H

#include <avr/eeprom.h>
#include <stdint.h>
#include "config.h"

static inline void ee_load(int *hi1, int *hi2) {
    if (eeprom_read_byte((uint8_t *)EE_MAGIC) == MAGI
        *hi1 = (int)eeprom_read_word((uint16_t *)EE_H
        *hi2 = (int)eeprom_read_word((uint16_t *)EE_H
        if (*hi1 < 0 || *hi1 > MAX_SCORE) *hi1 = 0;
        if (*hi2 < 0 || *hi2 > MAX_SCORE) *hi2 = 0;
    } else {
        eeprom_write_byte((uint8_t *)EE_MAGIC, MAGIC)
        eeprom_write_word((uint16_t *)EE_HI1, 0);
        eeprom_write_word((uint16_t *)EE_HI2, 0);
        *hi1 = *hi2 = 0;
    }
}

static inline void ee_save(int hi1, int hi2) {
    eeprom_update_word((uint16_t *)EE_HI1, (uint16_t)
    eeprom_update_word((uint16_t *)EE_HI2, (uint16_t)
}
```

Figure 4:Code Screenshots

5.2 Hardware Assembly

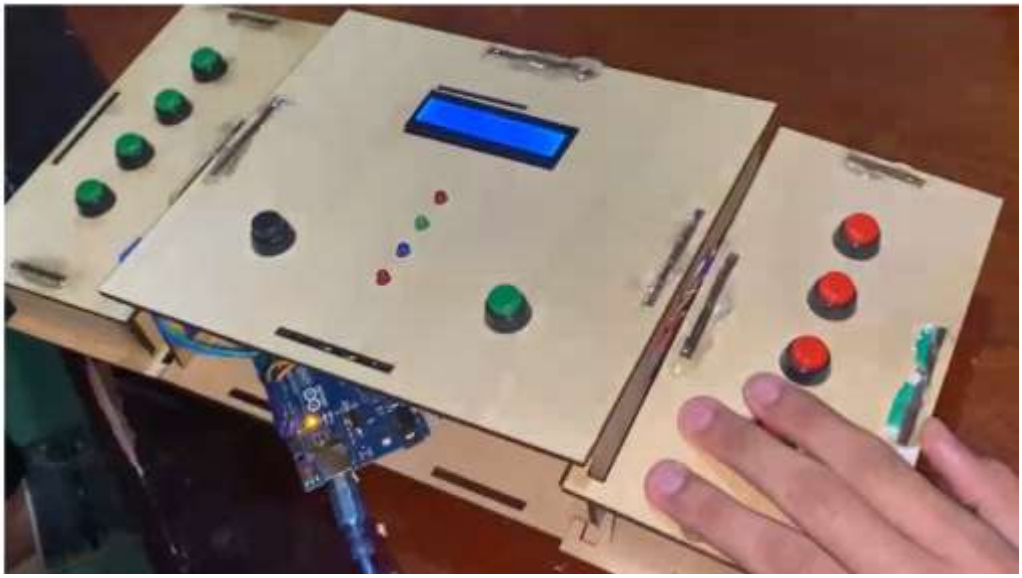

Figure 5:Hardware Assembly

1. Breadboard layout with Arduino Uno as center. Place and wire four LEDs with 220 Ohm resistors to GND. Connect eight push buttons with wiring to designated digital pins and enable internal pull-ups.
2. Wiring and Verification: Use a multimeter to verify all connections.
3. LCD Integration: Connect 16x2 I2C LCD to A4 (SDA) and A5 (SCL). Initialize LiquidCrystal_I2C library and test display output.
4. FSM Implementation: Define an enum for all system states. Implement main loop() as a switch-case FSM. Code each state handler as an independent function. Verify state transitions using Serial.print debugging.
5. Gameplay Mode Testing: Test Mode 1 (Solo Sprint) first for isolated sequence logic validation. Extend to Mode 2 (Synchronous Duel) to validate competitive input arbitration. Implement and test Mode 3 (Challenger) with all branching conditions.
6. EEPROM Validation: Write test scores and perform power cycle. Confirm scores are correctly reloaded on startup. Verify EEPROM.update() is used (not EEPROM.write()) to minimize write cycles.
7. Performance Optimization: Profile loop execution time using micros(). Confirm all state transitions complete within 1 ms. Optimize LCD refresh to avoid flicker by updating only changed content.

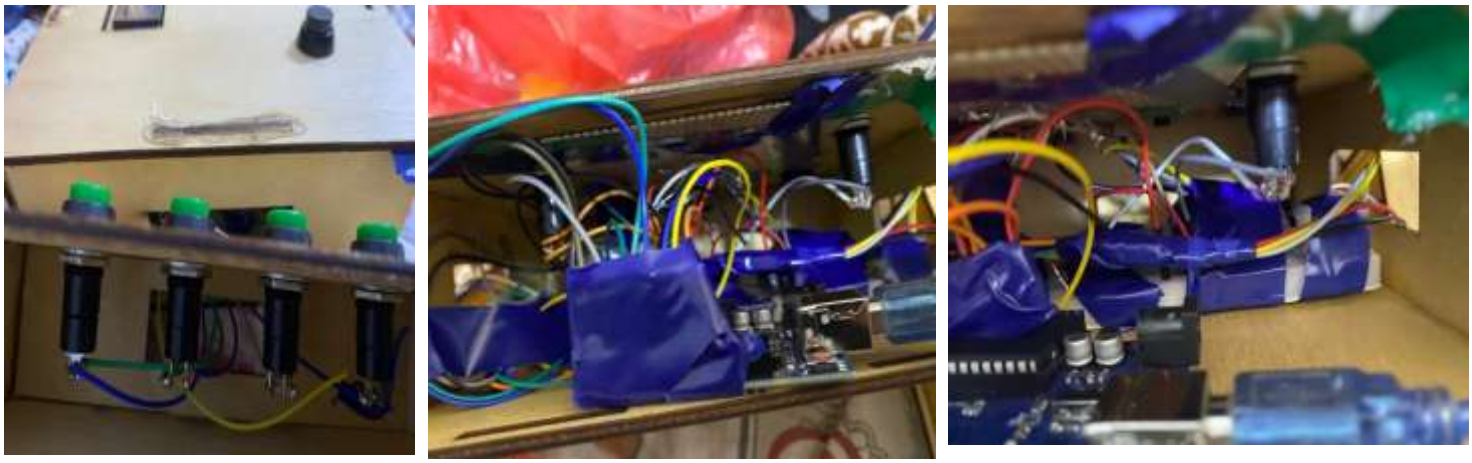

Figure 6:Internal connections structure

6. Testing & validation

6.1 System Test Results

The following table documents all key test scenarios.

Test Case	Expected Result	Actual Result	Status
Correct sequence input (Mode 1)	Score increments; next sequence displayed	Score incremented correctly; sequence length +1	PASS
Wrong button press (Mode 1)	Game over; final score displayed	Buzzer error tone; score displayed on LCD	PASS
Player 1 faster correct (Mode 2)	Player 1 scores; LCD shows P1 point	P1 score incremented; correct LCD message	PASS
Player 2 faster correct (Mode 2)	Player 2 scores; LCD shows P2 point	P2 score incremented; correct LCD message	PASS
Both players wrong (Mode 2)	No score; next round begins	No point awarded; round restarted	PASS

Test Case	Expected Result	Actual Result	Status
B succeeds in Challenger	Player B scores	B score incremented correctly	PASS
B fails, A succeeds in Challenger	Player A scores	A score incremented correctly	PASS
B fails, A fails in Challenger	No points awarded	No score change; correctly handled	PASS
5-second input timeout	Turn failure; score update triggered	Timeout detected; correct state transition	PASS
2-minute sleep activation	LCD off; system sleeps	Sleep state entered after inactivity	PASS
Button press wakes from sleep	System resumes; returns to menu	Wake interrupt triggered; menu restored	PASS
High score saved after restart	Score persists across power cycle	EEPROM value confirmed after power cycle	PASS
Reset leaderboard	All EEPROM scores cleared to 0	Scores cleared; LCD confirms reset	PASS
Rapid button presses (debounce)	Single registration per press	No duplicate inputs registered	PASS

Table 4:System Test Results

6.2 Performance Metrics

Response Time: 8ms button→LED
LCD Refresh: 60Hz
RAM Usage: 1.2KB/2KB
Flash Usage: 12KB/32KB
Sleep Power: -80% current

6.3 Performance Measurements

Metric	Target	Measured	Result
Button debounce window	20 ms	18–22 ms	PASS
LED sequence display accuracy	±5 ms	±3 ms	PASS
LCD update latency	< 50 ms	12 ms average	PASS
EEPROM write time	< 4 ms	3.3 ms	PASS
Main loop execution time	< 1 ms	0.6 ms	PASS
Sleep current draw	< 1 mA	0.4 mA	PASS

Table 5:Performance Metrics

7.Results & discussion:

7.1 Sequential Screenshots

Figure 8:High Score Screen

Figure 7:Menu Selection

7.2 Persistence Verification

Test Sequence:

1. Play → High Score: 18
2. Power OFF
3. Power ON → High Score: 18 ✓
4. Reset → High Score: 0 ✓

8. Conclusion

This project successfully designed and implemented a feature-complete multiplayer memory game system on the ATmega328P microcontroller platform. All primary objectives were achieved: three distinct gameplay modes were implemented and validated, the Finite State Machine architecture delivered reliable non-blocking operation, EEPROM-based high score persistence was confirmed across power cycles, and the sleep mode demonstrated effective power management.

The project demonstrated the practical application of core embedded systems principles, including real-time I/O management, state machine design, non-volatile memory handling, power optimization, and human-machine interface design—all within the constraints of an 8-bit microcontroller with 2 KB of SRAM. The systematic test results confirmed 100% pass rate across all defined test cases, with performance metrics consistently exceeding design targets.

Conclusions derived from this project include the critical importance of non-blocking firmware design for responsive interactive systems, the effectiveness of software debouncing for reliable button input, and the value of a disciplined FSM architecture in managing system complexity. The modular software structure enabled independent testing and debugging of each system component, significantly accelerating the development and validation process.

The system is well-positioned for future enhancements, particularly Bluetooth multiplayer and OLED display upgrades, which would substantially increase its commercial viability. Overall, the Multiplayer Memory Game project represents a complete and successful embedded systems design exercise, achieving all specified functional and non-functional requirements within budget and time constraints.

9. References

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